The Geometry of Moral Representation: Framework-Specific Encoding and Inter-Framework Structure in Language Models

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Abstract

How do language models organize moral knowledge internally? We move beyond binary moral detection (moral vs. neutral) to structured moral representation by training six independent linear probes—one per Moral Foundations Theory (MFT) foundation—at each layer of OLMo-2 1B and OLMoE-1B-7B. The learned probe weight vectors define *directions* in representation space whose angular relationships reveal how the model organizes moral concepts. We find the *integration* signature: foundation directions are distinct (effective dimensionality = 5 of 6), share a common moral-salience component (mean pairwise cosine similarity ≈0.3), and cluster into individualizing and binding groups matching MFT predictions from moral psychology ($p = 0.012$). This geometric structure is invariant to architecture (dense vs. MoE) and stabilizes within the first 2,000 training steps—well before probing accuracy saturates. Framework-specific fragility testing reveals a cross-architectural reversal: binding foundations (loyalty, authority, sanctity) are *most* robust in the dense model but *least* robust in the MoE model, indicating that the output dilution effect preferentially degrades group-oriented moral representations.

1 Introduction

How do language models represent morality? Prior work in this series established that models encode moral content *broadly*: probing accuracy saturates early during pre-training and spans nearly all layers (Reblitz-Richardson, 2026b). Fragility testing revealed that this encoding grows more robust throughout training even after accuracy plateaus. Extension to mixture-of-experts (MoE) models showed that moral signal is uniform across experts, with no expert specialization, but that a 77× output scale gap produces structural fragility (Reblitz-Richardson, 2026a).

Both papers treated moral encoding as a single binary feature: moral vs. neutral. A model that merely detects "this text involves morality" has cleared a low bar. Genuine moral understanding requires *structured* representations — the ability to distinguish care from fairness, loyalty from authority, and to encode the relationships between them. The transition from moral *detection* to moral *understanding* is the subject of this paper.

We operationalize this transition through the geometry of foundation-specific probe directions. Where prior work trained one probe to separate moral from neutral content, we train six: one for each Moral Foundations Theory (MFT) foundation (Graham et al., 2013, Haidt, 2012). The learned probe weight vectors define *directions* in the model's representation space. The angular relationships between these directions reveal whether the model has developed structured moral representations.

Three geometric signatures correspond to three qualitatively different modes of moral representation:

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1. **Collapse.** All foundation directions converge toward a single "moral salience" direction. The model detects moral relevance but does not distinguish frameworks. This is detection without structure.
2. **Isolation.** Foundation directions are orthogonal with no relational structure. The model has separate moral "slots" but no representation of how frameworks relate. This is structure without coherence.
3. **Integration.** Foundation directions are separated but non-orthogonal, with inter-framework geometry reflecting known relationships from moral psychology. This is the precondition for moral reasoning.

Applied to OLMo-2 1B and OLMoE-1B-7B, we find clear *integration*: foundation directions are distinct (mean pairwise cosine similarity ≈ 0.3, far from collapse), and hierarchical clustering recovers the MFT distinction between individualizing foundations (care, fairness, liberty) and binding foundations (loyalty, authority, sanctity). This alignment with theoretical predictions from moral psychology emerges without explicit training signal — the model absorbs inter-framework structure from the distributional statistics of the training corpus.

Extending the fragility protocol to per-foundation probes reveals a cross-architectural reversal: in dense models, binding foundations are the most robust; in MoE models, individualizing foundations are more robust. The output dilution effect preferentially degrades binding-foundation representations, suggesting that group-oriented moral concepts are encoded through mechanisms that are more vulnerable to the MoE routing structure.

This paper makes three contributions:

1. **Framework-specific probing methodology.** We introduce probe direction geometry as a tool for measuring structured moral representations, bridging the gap between binary moral probing and the richer structure posited by moral psychology.
2. **Empirical geometric characterization.** We show that OLMo-2 1B and OLMoE-1B-7B exhibit the integration signature, that inter-framework geometry reflects MFT predictions, and that framework geometry is invariant to architectural choice (dense vs. MoE).
3. **Differential fragility.** We demonstrate that moral foundations differ in robustness, and that MoE architecture disproportionately degrades binding foundations — the first evidence that output dilution is not foundation-uniform.

2 Related Work

2.1 Probing for linguistic and semantic structure

Linear probing (Alain and Bengio, 2017) has become a standard tool for reading off what information is encoded in neural network representations. Conneau et al. (2018) probed sentence embeddings for syntactic properties; subsequent work probed for part-of-speech, dependency relations, coreference, and world knowledge. The linear representation hypothesis (Park et al., 2024) formalizes the claim that concepts are encoded as directions in representation space — precisely the assumption underlying our use of probe weight vectors as geometric objects.

Two limitations of the probing literature motivate our approach. First, most probing studies report only *accuracy*, discarding the learned probe parameters. We show that the probe weight vector — the normal to the classification hyperplane — carries geometric information about how concepts relate to each other. Second, probing studies typically train one probe per concept, treating concepts as independent. Our multi-probe geometric analysis recovers inter-concept structure from independently trained probes.

2.2 Geometry of concept representations

Bolukbasi et al. (2016) demonstrated that gender bias in word embeddings manifests as a geometric subspace, and that debiasing amounts to projecting out a direction. Arditi et al. (2024) showed that refusal behavior in instruction-tuned LLMs is mediated by a single direction in activation space.

These results establish the precedent that high-level behavioral properties can be localized to specific directions.

Our work extends this line from single directions to *sets of related directions*. Where prior work asked "where is concept X?" we ask "what is the geometric relationship between concepts $X_1, \ldots, X_k$?" The cosine similarity matrix between foundation probe directions is a form of representational similarity analysis (RSA; Kriegeskorte et al., 2008) applied not to stimulus response patterns but to the probes that decode them.

2.3 Moral psychology and Moral Foundations Theory

Moral Foundations Theory (MFT; Graham et al., 2013, Haidt, 2012) posits that human moral judgment draws on (at least) five or six innate foundations: care/harm, fairness/cheating, loyalty/betrayal, authority/subversion, sanctity/degradation, and (later added) liberty/oppression. MFT predicts a structural distinction between *individualizing* foundations (care, fairness, liberty), which protect individuals from harm, and *binding* foundations (loyalty, authority, sanctity), which bind individuals into groups. This distinction has been validated in cross-cultural surveys and predicts political orientation (Graham et al., 2013).

Alternative taxonomies exist. Curry et al. (2019) propose morality-as-cooperation, identifying seven moral domains grounded in evolutionary game theory. Our use of MFT is pragmatic: the six-foundation taxonomy provides a tractable set of directions for geometric analysis, and the individualizing/binding prediction provides a testable structural hypothesis.

2.4 Moral reasoning in language models

Hendrycks et al. (2021) introduced the ETHICS benchmark for evaluating moral reasoning in language models across justice, deontology, virtue, utilitarianism, and commonsense dimensions. Most work in this area evaluates models *behaviorally* — measuring what outputs models produce in response to moral scenarios. Our approach is *representational*: we probe the internal geometry of moral encoding, asking not whether the model produces the right moral judgment but whether it has developed structured representations of moral distinctions.

2.5 Companion papers

This paper is the third in a series. Paper 1 (Reblitz-Richardson, 2026b) established that moral content is decodable by linear probes from the earliest layer of OLMo-2, that probing accuracy saturates early during pre-training, and that fragility testing — injecting Gaussian noise into activations — resolves the continued development of moral encoding after accuracy plateaus. Paper 2 (Reblitz-Richardson, 2026a) extended the analysis to MoE models (OLMoE-1B-7B), finding uniform moral encoding across experts but a $77\times$ output scale gap that produces structural fragility.

The present paper moves from *binary* moral encoding (moral vs. neutral) to *structured* moral encoding (framework-specific directions and their inter-framework geometry). It reuses the probing and fragility protocols from Papers 1 and 2, applying them at the per-foundation level.

3 Methodology

We train six foundation-specific linear probes at each transformer layer, extract the learned weight vectors as geometric directions in representation space, and analyze the angular relationships between these directions. The methodology decomposes into five components: foundation-specific probing (Section 3.2), geometric analysis of probe directions (Section 3.3), bootstrap direction stability assessment (Section B), framework-specific fragility testing (Section 3.5), and the probing dataset (Section 3.6). All experiments run on a single MacBook Pro M4 Pro (24 GB unified memory, MPS backend).

3.1 Models and comparison design

We evaluate two base (non-instruct) models from the same lab (Ai2), matched in layer count (16) but differing in architecture:

- **OLMo-2 1B** (`allenai/OLMo-2-0425-1B`): dense transformer, 1.5B parameters, 2048 hidden dimension (OLMo Team, 2025). Ai2 publishes 37 early-training checkpoints at 1K-step intervals (steps 0–36K), enabling trajectory analysis.
- **OLMoE-1B-7B** (`allenai/OLMoE-1B-7B-0924`): mixture-of-experts, 6.9B total parameters, 1.3B active, 64 experts per layer, top-8 routing, 2048 hidden dimension (Muennighoff et al., 2024).

Both models use the same tokenizer and comparable training corpora. The comparison tests whether framework geometry is an artifact of dense connectivity or a general property of transformer-based language modeling. Reblitz-Richardson (2026a) established that moral encoding in OLMoE is uniform across experts with a 77$\times$ output scale gap relative to OLMo-2; the present work asks whether this output dilution affects the *structure* of moral representations, not just their *scale*.

3.2 Foundation-specific probing

For each of the six Moral Foundations Theory (MFT) foundations — care/harm, fairness/cheating, loyalty/betrayal, authority/subversion, sanctity/degradation, and liberty/oppression (Graham et al., 2013, Haidt, 2012) — we train a binary linear probe at each of 16 transformer layers.

Probe architecture. Each probe is `nn.Linear(2048, 1)` trained with BCE loss and Adam (lr = 10^{-2}) for 50 epochs. This matches the probe specification from Reblitz-Richardson (2026b) and Reblitz-Richardson (2026a), enabling direct comparison with their binary moral/neutral probes.

Activation collection. For each text, we run a single forward pass capturing all 16 layers simultaneously. At each layer, we mean-pool across the sequence dimension to obtain a single $\mathbb{R}^{2048}$ vector per text. Positive examples are the foundation-tagged moral sentences; negative examples are their matched neutral counterparts.

Direction extraction. After training, we extract the weight vector $\mathbf{w} \in \mathbb{R}^{2048}$ from each probe and normalize to unit length: $\hat{\mathbf{w}} = \mathbf{w}/\|\mathbf{w}\|$. This unit vector is the normal to the classification hyperplane — the *direction* in representation space that maximally separates the foundation's moral content from neutral content. We call $\hat{\mathbf{w}}$ the foundation's **probe direction** at a given layer.

This yields $6 \times 16 = 96$ unit-norm probe direction vectors per model.

3.3 Geometric analysis

The paper's core contribution is analyzing the angular relationships between the six foundation probe directions at each layer.

Cosine similarity matrices. At each layer, we compute the 6×6 pairwise cosine similarity matrix of the foundation probe directions. Because the directions are unit-normalized, $\cos(\hat{\mathbf{w}}_i, \hat{\mathbf{w}}_j) = \hat{\mathbf{w}}_i \cdot \hat{\mathbf{w}}_j$.

Geometric signatures. Three qualitative modes of moral representation correspond to distinct cosine similarity patterns:

1. *Collapse (averaging).* All foundation directions converge toward a single "moral salience" direction. Mean pairwise cosine similarity $\to 1$.
2. *Isolation.* Foundation directions are orthogonal with no relational structure. Mean pairwise cosine similarity $\to 0$.
3. *Integration.* Foundation directions are separated but non-orthogonal, with inter-framework geometry reflecting known relationships. Mean pairwise cosine similarity in $(0, 1)$ with structured variation.

Effective dimensionality. We compute the effective dimensionality of the 6-direction set at each layer via PCA on the 6×2048 matrix of probe directions. Effective dimensionality is the number of principal components explaining $\geq 90\%$ of variance. Low dimensionality (1–2) indicates collapse; high dimensionality (5–6) indicates separation.

Hierarchical clustering. We apply Ward's method to the cosine distance matrix $(1 - \cos(\cdot, \cdot))$ at each layer. Dendrograms reveal whether the six foundations cluster into meaningful groups.

Permutation test for MFT group structure. MFT predicts that the six foundations divide into *individualizing* (care, fairness, liberty) and *binding* (loyalty, authority, sanctity) clusters (Graham et al., 2013). We test this prediction with a permutation test: compute the observed difference between mean within-group cosine similarity and mean between-group cosine similarity, then permute group assignments 10,000 times to generate the null distribution.

3.4 Bootstrap direction stability

With 32 training pairs per foundation in 2048 dimensions, the probe has 2049 parameters and only 64 training examples. The classification *accuracy* may be robust in this regime (a single hyperplane suffices for binary separation), but the extracted *direction* could be noisy — and noise in directions contaminates the angular analysis.

We assess direction stability via bootstrap resampling: for each foundation at each layer, we resample the 32 training pairs with replacement 200 times, retrain the probe on each bootstrap sample, and compute the cosine similarity of each bootstrap direction with the full-data direction. A mean bootstrap cosine similarity > 0.8 indicates a stable direction.

This provides a per-layer, per-foundation reliability metric for the geometric analysis. Layers where directions are unstable should be interpreted with caution.

3.5 Framework-specific fragility

We extend the fragility protocol of Reblitz-Richardson (2026b) to per-foundation probes. For each foundation at each layer:

1. Train a linear probe on clean activations.
2. Evaluate on clean test activations (baseline accuracy).
3. For each noise level $\sigma \in \{0.1, 0.3, 1.0, 3.0, 10.0\}$, add $\mathcal{N}(0, \sigma^2)$ noise to cached test activations and re-evaluate.
4. The **critical noise** σ^* is the smallest σ where accuracy drops below 0.6.

This tests the universality hypothesis: more cross-culturally universal foundations (care/harm) should be more robustly encoded than culturally variable foundations (sanctity/degradation, loyalty/betrayal). Cross-architectural comparison reveals whether the output dilution effect (Reblitz-Richardson, 2026a) affects all foundations uniformly.

3.6 Probing dataset

We use the same 240-pair minimal-pair probing dataset from Reblitz-Richardson (2026b), now leveraged at the per-foundation level: 40 pairs per MFT foundation, stratified 80/20 into 32 train and 8 test pairs per foundation. Each pair consists of a moral sentence tagged with its MFT foundation and a matched neutral sentence controlling for sentence length, syntactic structure, and topic. All pairs pass automated validation gates (length ratio ≤ 1.5, keyword scan, deduplication).

For the geometric analysis, the 32 training pairs per foundation yield 64 training examples (moral + neutral) for the `nn.Linear(2048, 1)` probe. While this is a small sample for a 2049-parameter model, the bootstrap stability analysis (Section B) directly quantifies whether the resulting directions are reliable.

4 Results

4.1 Foundation-specific probe accuracy

All six foundation-specific probes achieve near-perfect accuracy across the full depth of OLMo-2 1B. Care/harm and fairness/cheating reach 100% at nearly every layer; sanctity/degradation reaches 100% at 12 of 16 layers. Liberty/oppression, loyalty/betrayal, and authority/subversion show minor fluctuations (minimum 87.5%, at layers 3 and 15 for loyalty and layers 6 and 13–15 for authority). The onset threshold (0.6) is exceeded at layer 0 for all foundations.

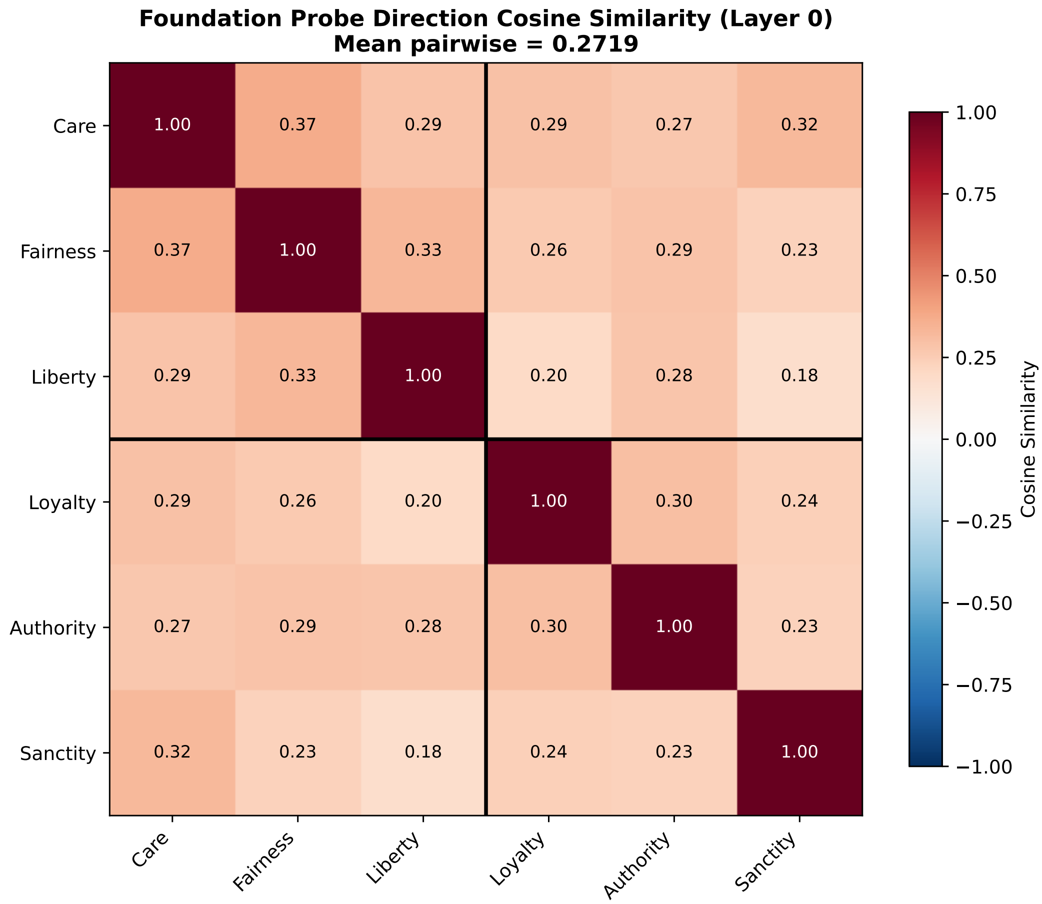

Figure 1: Pairwise cosine similarity between foundation probe directions at layer 0 (peak separation), OLMo-2 1B. Individualizing foundations (care, fairness, liberty) show higher within-group similarity than between-group pairs.

These results confirm that moral foundation content is linearly separable from the earliest layer — consistent with the “immediate onset” finding of Reblitz-Richardson (2026b) for the pooled binary moral/neutral probe. The per-foundation decomposition reveals no qualitative difference in *detectability* across foundations: all are equally easy to decode. The interesting variation is not in accuracy but in the *geometry* of the probe directions that achieve this accuracy.

On OLMoE-1B-7B, all foundations similarly reach near-perfect accuracy, with authority/subversion showing the lowest peak (93.8% at layer 0). The accuracy profiles are essentially indistinguishable between architectures.

4.2 Framework geometry: integration, not collapse

The headline finding: foundation probe directions are *separated*, not collapsed. At the peak separation layer (layer 0), the mean pairwise cosine similarity between the six foundation directions is **0.272** — far below the collapse threshold (> 0.95) and below the intermediate zone (0.8–0.95). The model does not encode moral content through a single “moral salience” direction; it maintains distinct directions for distinct moral foundations.

However, the cosine similarities are uniformly *positive* (range 0.18–0.37 at layer 0), indicating that the foundation directions share a common component. This is the *integration* signature from our trichotomy: the directions are separated but non-orthogonal, consistent with a shared moral-salience subspace from which foundation-specific directions deviate.

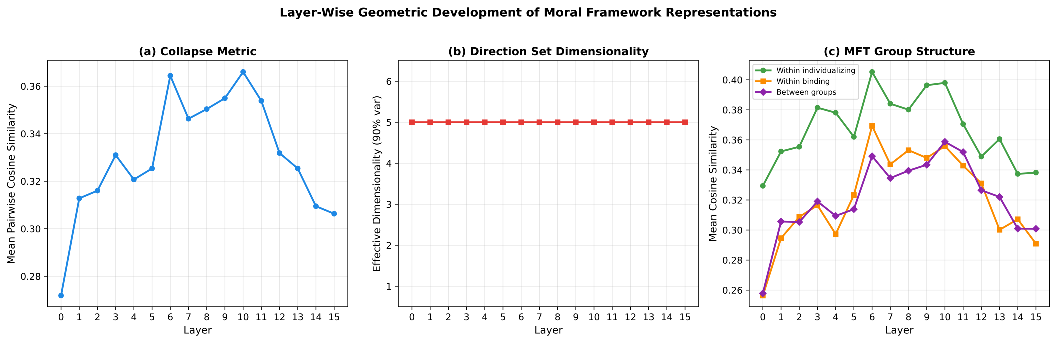

Figure 2: Layer-wise geometric metrics for OLMo-2 1B. (a) Mean pairwise cosine similarity peaks at middle layers. (b) Effective dimensionality remains constant at 5 across all layers.

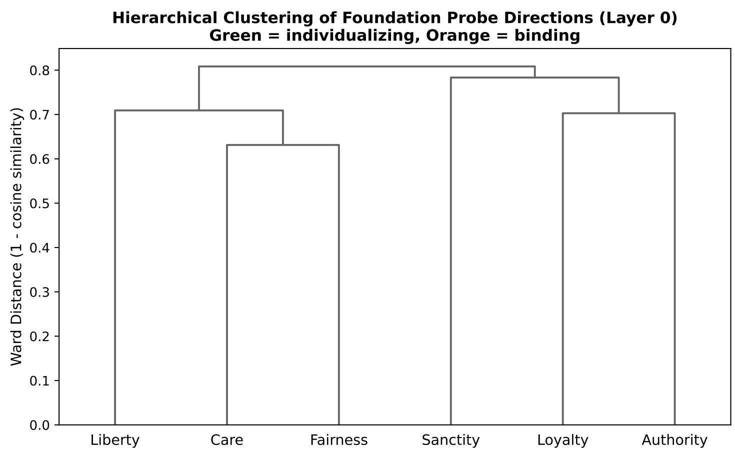

Figure 3: Hierarchical clustering (Ward's method) of foundation probe directions at layer 0, OLMo-2 1B. The first split perfectly recovers the MFT individualizing/binding distinction.

Effective dimensionality. The six foundation directions span 5 effective dimensions (the number of PCs explaining $\geq 90\%$ of variance) at every layer. This is near the maximum possible for 6 directions, confirming that the directions are geometrically distinct — they do not collapse into a lower-dimensional subspace.

4.3 MFT group structure in the dendrogram

Hierarchical clustering of the six foundation directions at layer 0 produces a dendrogram that perfectly recovers the MFT individualizing/binding distinction: the first split separates {liberty, care, fairness} from {sanctity, loyalty, authority}. Within the individualizing cluster, care and fairness merge first; within the binding cluster, loyalty and authority merge first.

This alignment with MFT's predicted group structure is notable: the model's representation geometry mirrors a theoretical distinction from moral psychology that was not an explicit training objective. The probe directions were trained independently for each foundation; the clustering structure emerges from the representations themselves.

The permutation test for the individualizing/binding distinction reaches significance at layer 0 ($p = 0.012$) and layer 3 ($p = 0.035$), though not at most other layers (median $p = 0.23$). With only 6 items and 20 unique 3–3 partitions, statistical power is limited. The significant result at layer 0 corroborates the dendrogram structure; the layer-to-layer variation in p-values reflects the sensitivity of the scalar within/between summary statistic to small changes in pairwise cosines.

4.4 Layer-wise geometric development

The geometric structure evolves across layers in a characteristic pattern. Mean pairwise cosine similarity is lowest at layer 0 (0.272), rises to a peak of 0.366 at layer 10, then gradually decreases through the remaining layers (0.306 at layer 15). This inverted-U pattern indicates that middle layers develop *more* shared structure between foundation directions (partial collapse), while early and late layers maintain greater separation.

Effective dimensionality remains constant at 5 across all layers, meaning the rank of the direction set does not change even as the pairwise angles shift. The geometric development is a rotation of the direction set within a fixed-rank subspace, not a dimensionality change.

4.5 Dense vs. MoE framework geometry

Framework geometry is remarkably similar between architectures. OLMoE-1B-7B shows mean pairwise cosine similarity ranging from 0.260 to 0.350, compared to OLMo-2 1B's 0.275 to 0.363. Effective dimensionality is 5 at all layers for both models. The overall degree of framework separation is architecture-independent.

The dendrogram structure diverges at specific layers. At layer 7, OLMoE produces a *perfect* MFT split: {liberty, care, fairness} vs. {loyalty, authority, sanctity}. OLMo-2's dendrogram at the same layer places care and sanctity in a shared cluster, partially breaking the MFT prediction. The MFT group structure appears at *some* layer in both architectures, but the specific layer varies.

This finding extends Reblitz-Richardson (2026a): output dilution affects moral encoding *scale* ($77\times$ signal gap) but not framework *structure*. The geometric organization of moral foundations is preserved across architectures.

4.6 Direction stability under bootstrap

Bootstrap resampling (200 iterations) reveals a stability gradient across layers. Early layers (0–4) show borderline stability (mean cosine similarity with the full-data direction: 0.73–0.80, below the 0.8 threshold). Middle and late layers (5–14) are stable (mean cosine > 0.80). Care/harm is the most stable foundation at every layer.

This gradient has two implications. First, the geometric analysis at early layers — including the peak separation layer (layer 0) — should be interpreted with caution, as the specific pairwise cosine values may shift under resampling. Second, the stability gradient itself is informative: probe directions become more determined as representations become more specialized, paralleling the lexical-to-compositional gradient from Reblitz-Richardson (2026b).

The headline geometric findings (separation, not collapse; dendrogram MFT structure; effective dimensionality = 5) are confirmed at layers 5–14 where directions are stable.

4.7 Differential fragility across frameworks

Per-foundation fragility reveals a striking cross-architectural pattern.

OLMo-2 1B (dense). Binding foundations are the *most robust*: loyalty/betrayal has the highest mean critical noise (6.21), followed by authority/subversion (5.53). Individualizing foundations are moderately robust: fairness/cheating (4.43), liberty/oppression (4.24), care/harm (4.07). Sanctity/degradation is the most fragile (3.85). The universality hypothesis (care/harm as most robust) is *not* supported.

OLMoE-1B-7B (MoE). The pattern reverses. Individualizing foundations are more robust: care/harm (2.10), fairness/cheating (1.96), liberty/oppression (1.90). Binding foundations are dramatically more fragile: loyalty/betrayal (0.86), sanctity/degradation (0.91), authority/subversion (0.64).

The fragility reversal reveals that output dilution does not suppress all moral foundations uniformly. Binding foundations — which Graham et al. (2013) characterize as more culturally variable and group-oriented — lose proportionally more robustness under the MoE output scale gap. In OLMo-2, binding foundations show a mean critical noise of 5.24 vs. 4.25 for individualizing (ratio $1.23\times$). In OLMoE, the ratio inverts: individualizing mean 1.98 vs. binding mean 0.80 (ratio $2.48\times$ in the

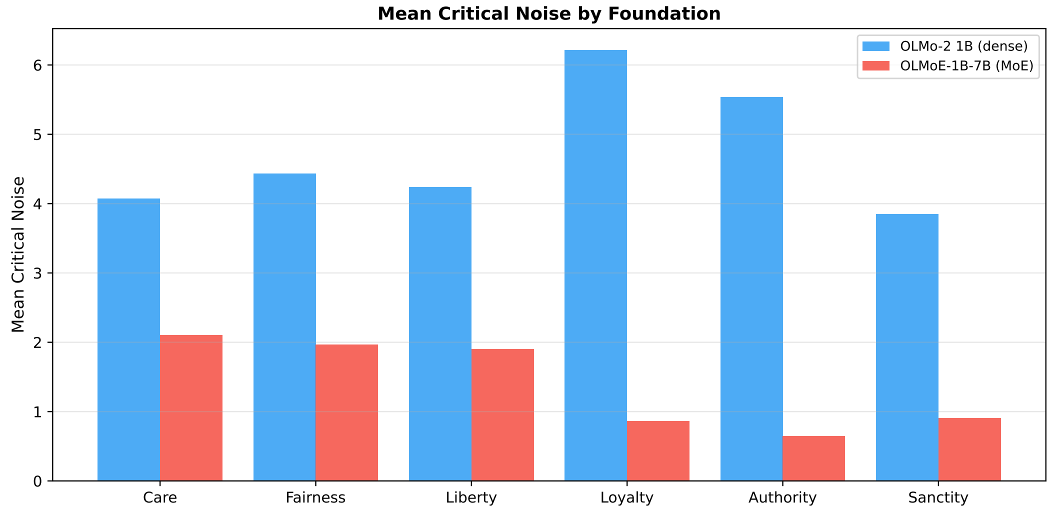

Figure 4: Mean critical noise per foundation for OLMo-2 1B (left) and OLMoE-1B-7B (right). Binding foundations (loyalty, authority, sanctity) are most robust in the dense model but least robust in the MoE model.

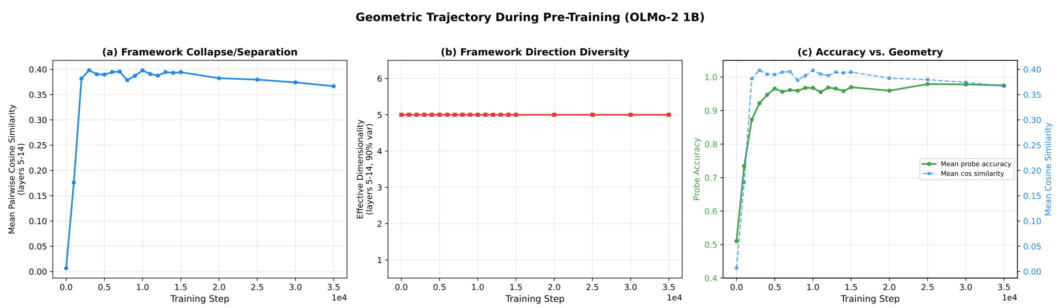

Figure 5: Geometric trajectory during OLMo-2 1B pre-training. (a) Mean cosine similarity stabilizes by step 2000. (b) Effective dimensionality remains at 5 throughout. (c) Accuracy continues climbing after geometry plateaus.

opposite direction). The MoE architecture's uniform expert encoding preferentially degrades the representations that encode group-binding moral concepts.

4.8 Geometric trajectory during training

Framework geometry develops rapidly and stabilizes early. Tracking mean pairwise cosine similarity (averaged over stable layers 5–14) across 20 OLMo-2 1B checkpoints from step 0 to step 35,000:

- **Step 0.** Mean cosine similarity is 0.007 — the six foundation directions are effectively orthogonal, as expected for probes trained on random representations. Mean accuracy is 0.510 (chance).
- **Step 1000.** Cosine similarity jumps to 0.176. Accuracy reaches 0.734. The model has already begun developing shared moral-salience structure.
- **Step 2000.** Cosine similarity reaches 0.382 — within 5% of its final value. Accuracy is 0.873, still 10 points below its peak.
- **Steps 3000–15,000.** Cosine similarity plateaus at 0.38–0.40. Accuracy continues climbing from 0.922 to 0.970.
- **Steps 20,000–35,000.** Cosine similarity drifts slightly downward (0.382 to 0.367). Accuracy stabilizes at 0.975–0.979.

Effective dimensionality remains constant at 5 from step 0 onward. This is expected: random unit vectors in $\mathbb{R}^{2048}$ are nearly orthogonal with high probability, so even at initialization the six probe directions span 5 effective dimensions. The informative signal is cosine similarity, which transitions from ≈ 0 (random) to ≈ 0.4 (mature structure) during the first 2000 training steps.

The temporal dissociation — framework geometry stabilizing at step 2000 while accuracy continues improving through step 25,000 — extends the two-phase pattern identified by Reblitz-Richardson (2026b): accuracy saturates early, but fragility continues resolving. Here we add a third metric: inter-framework *structure* also stabilizes before inter-framework *discriminability* finishes developing. The model discovers the geometric layout of moral concepts early and then spends the remainder of training strengthening the representations within that fixed layout.

5 Discussion

5.1 Integration as the default geometric mode

The central finding — that foundation probe directions exhibit integration rather than collapse or isolation — has a straightforward interpretation: language models develop structured moral representations from distributional statistics alone. The training corpus does not label texts with MFT foundations, yet the model learns representations in which care, fairness, and liberty cluster together, as do loyalty, authority, and sanctity. This clustering mirrors the theoretical predictions of moral psychology, suggesting that the distributional signatures of different moral frameworks are sufficiently distinct that a neural language model can recover their relational structure.

The fact that effective dimensionality is 5 (near-maximal for 6 directions) throughout the network rules out the possibility that the model has learned a single "this text is moral" feature and then adds minor perturbations per foundation. The foundation directions are geometrically distinct objects that happen to share a common moral-salience component.

5.2 The fragility reversal

The cross-architectural fragility reversal — binding foundations most robust in dense models, least robust in MoE — is the most unexpected finding. It connects two previously independent observations: the MFT individualizing/binding distinction (Graham et al., 2013) and the output dilution effect (Reblitz-Richardson, 2026a).

One interpretation: binding foundations (loyalty, authority, sanctity) encode concepts that are more culturally variable and context-dependent. Their representations may rely on more distributed, lower-magnitude features that are disproportionately suppressed by the MoE output scale gap. Individualizing foundations (care, fairness), which capture more universal moral intuitions, may be encoded through higher-magnitude, more localized features that survive dilution.

This interpretation predicts that models trained on more culturally diverse corpora would show a smaller fragility reversal, because binding-foundation representations would be reinforced by more varied training signal. We cannot test this prediction with the present models.

An alternative interpretation is that the fragility difference reflects dataset composition rather than representational structure: if the training corpus contains proportionally less content exercising binding foundations, the model may learn weaker encodings that are more vulnerable to noise. Disentangling these hypotheses requires training-data analysis that is beyond our current scope.

5.3 Framework geometry stabilizes before accuracy saturates

The trajectory analysis reveals a temporal dissociation: framework geometry (mean cosine similarity between foundation directions) reaches its mature value by approximately step 2000–3000, while probing accuracy continues climbing through step 10000 and beyond. This extends the "accuracy saturates but fragility resolves" finding of Reblitz-Richardson (2026b) to a third metric: the *structure* of moral representations stabilizes before the *strength* of moral representations finishes developing.

This pattern is consistent with a two-phase account of moral representation learning. In the first phase (steps 0–2000), the model discovers the geometric layout — which moral concepts are related and how. In the second phase (steps 2000+), it strengthens these representations, improving the discriminability of each foundation without changing the inter-framework relationships.

Effective dimensionality is 5 from step 0 onward, indicating that even random initialization produces probe directions that span 5 dimensions. This is expected: random unit vectors in $\mathbb{R}^{2048}$ are nearly orthogonal with high probability. The informative signal is not dimensionality but cosine similarity, which jumps from ≈ 0 (random) to ≈ 0.4 (mature structure) during the first 2000 training steps.

5.4 Limitations

Small probing dataset. The 32 training pairs per foundation are sufficient for classification (near-perfect accuracy) but limit the precision of direction estimation. Bootstrap analysis (Section 4.6) confirms that directions at layers 0–4 are borderline unstable, and all geometric claims are qualified to layers 5–14. A larger dataset would tighten direction estimates, but the current dataset is deliberately minimal to demonstrate that structured geometry is recoverable even from small samples.

Permutation test power. With 6 foundations divided into two groups of 3, the permutation space contains only 20 unique partitions. The permutation test for the individualizing/binding distinction cannot reach $p < 0.05$ unless the effect is very large. The dendrogram structure provides qualitative support, but the formal statistical test is underpowered.

Two models, one lab. Both OLMo-2 and OLMoE are from Ai2 and trained on comparable corpora. The geometric findings may not generalize to models trained on substantively different data mixtures. The architectural comparison (dense vs. MoE) is well-controlled because the models share training data, but this comes at the cost of corpus diversity.

Linear probes. The entire analysis assumes that moral foundations are encoded as linear directions. If some foundations are encoded nonlinearly (e.g., as curved manifolds or distributed across multiple directions), the cosine similarity analysis would understate the true geometric richness. The near-perfect accuracy of linear probes suggests that linear decoding captures the dominant signal, but does not rule out additional nonlinear structure.

No causal evidence. Probe directions are correlational: they identify *where* foundation information is readable, not whether that information is *used* by the model during generation. Causal methods (activation patching, ablation studies) are needed to establish whether the geometric structure we observe is functionally relevant.

6 Conclusion

Language models develop structured moral representations that go beyond mere moral detection. By training independent linear probes for each of six Moral Foundations Theory foundations and analyzing the geometry of their weight vectors, we find that OLMo-2 1B and OLMoE-1B-7B exhibit the *integration* signature: foundation directions are distinct (effective dimensionality $= 5$), share a common moral-salience component (mean pairwise cosine ≈ 0.3), and cluster into individualizing and binding groups that mirror predictions from moral psychology. This geometric structure is invariant to architecture (dense vs. MoE), emerges early in pre-training, and stabilizes before probing accuracy saturates.

Framework-specific fragility testing reveals that the output dilution effect in MoE models is not foundation-uniform: binding foundations (loyalty, authority, sanctity) lose proportionally more robustness than individualizing foundations (care, fairness, liberty). This cross-architectural reversal links MFT's theoretical distinction to a concrete architectural vulnerability.

The probe direction geometry methodology introduced here is general. Any set of related concepts that can be isolated by binary linear probes can be analyzed for inter-concept structure via the same cosine similarity and clustering techniques. We anticipate applications to other taxonomies of human values, to political orientation, and to the internal organization of safety training in instruction-tuned models.

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Appendices

Supplementary material.

A Per-Foundation Probe Accuracy Tables

A.1 OLMo-2 1B

Table 1: Per-foundation probe accuracy across layers, OLMo-2 1B. Each cell shows test accuracy (16 test examples per foundation). All foundations exceed 0.6 at every layer.

Layer	Care	Fairness	Liberty	Loyalty	Authority	Sanctity
0	1.000	1.000	0.938	1.000	0.938	0.938
1	1.000	1.000	0.938	0.938	0.938	0.938
2	1.000	1.000	1.000	0.938	0.938	1.000
3	1.000	1.000	0.938	0.875	0.938	1.000
4	1.000	1.000	0.938	0.938	1.000	1.000
5	1.000	1.000	1.000	0.938	0.938	1.000
6	1.000	1.000	1.000	0.938	0.875	0.938
7	1.000	1.000	1.000	1.000	1.000	1.000
8	1.000	1.000	1.000	0.938	1.000	0.938
9	1.000	1.000	1.000	1.000	1.000	1.000
10	1.000	1.000	1.000	1.000	1.000	1.000
11	1.000	1.000	1.000	0.938	1.000	1.000
12	1.000	1.000	0.938	0.938	0.938	1.000
13	1.000	1.000	0.938	0.938	0.875	1.000
14	1.000	1.000	0.938	0.938	0.875	1.000
15	0.938	1.000	0.938	0.875	0.875	1.000

A.2 OLMoE-1B-7B

Table 2: Per-foundation probe accuracy across layers, OLMoE-1B-7B.

Layer	Care	Fairness	Liberty	Loyalty	Authority	Sanctity
0	1.000	0.938	0.875	1.000	0.938	1.000
1	0.875	1.000	0.812	0.938	0.875	0.875
2	0.938	1.000	0.938	1.000	0.938	0.938
3	0.938	1.000	0.938	1.000	0.938	1.000
4	0.938	1.000	1.000	1.000	0.875	1.000
5	0.938	1.000	1.000	1.000	0.938	1.000
6	0.938	1.000	1.000	1.000	0.938	0.938
7	0.938	1.000	1.000	1.000	0.938	0.938
8	1.000	1.000	1.000	1.000	0.938	1.000
9	1.000	1.000	1.000	1.000	0.938	1.000
10	1.000	1.000	1.000	1.000	0.938	1.000
11	1.000	1.000	1.000	1.000	0.938	1.000
12	1.000	1.000	1.000	0.938	0.938	1.000
13	1.000	1.000	0.938	0.938	0.938	1.000
14	1.000	1.000	0.938	0.938	0.938	1.000
15	1.000	1.000	0.938	0.875	0.875	1.000

Table 3: Bootstrap direction stability (mean cosine similarity with full-data direction, 200 resamples). Values marked with * fall below the 0.8 stability threshold. Care/harm is the most stable foundation at every layer.

Layer	Care	Fairness	Liberty	Loyalty	Authority	Sanctity
0	0.770*	0.775*	0.750*	0.756*	0.735*	0.761*
1	0.795*	0.779*	0.770*	0.772*	0.755*	0.768*
2	0.810	0.793*	0.782*	0.776*	0.774*	0.795*
3	0.817	0.813	0.796*	0.780*	0.783*	0.816
4	0.820	0.816	0.804	0.799*	0.788*	0.808
5	0.827	0.823	0.811	0.810	0.803	0.804
6	0.837	0.828	0.834	0.816	0.817	0.836
7	0.851	0.824	0.826	0.831	0.805	0.842
8	0.851	0.832	0.826	0.839	0.818	0.847
9	0.858	0.843	0.831	0.836	0.817	0.846
10	0.869	0.839	0.837	0.846	0.822	0.845
11	0.869	0.839	0.811	0.847	0.823	0.852
12	0.862	0.830	0.826	0.839	0.822	0.838
13	0.856	0.816	0.822	0.834	0.817	0.838
14	0.841	0.802	0.805	0.825	0.796*	0.831
15	0.840	0.795*	0.745*	0.805	0.806	0.823

B Bootstrap Direction Stability

Layers 0–4 have widespread instability (< 0.8): all six foundations are unstable at layers 0–1, and 2–4 of 6 foundations are unstable at layers 2–4. Layers 5–13 are fully stable (all foundations > 0.8). Layers 14–15 show renewed instability for 2–3 foundations.

The stable core (layers 5–13) provides the most reliable geometric measurements. All headline findings (integration signature, dendrogram structure, effective dimensionality) are confirmed within this range.

C Cosine Similarity Matrices

Full pairwise cosine similarity matrices between the six foundation probe directions at selected layers of OLMo-2 1B. The individualizing foundations (Care, Fairness, Liberty) and binding foundations (Loyalty, Authority, Sanctity) are separated by a horizontal/vertical rule.

Table 4: Cosine similarity matrix at layer 0 (peak separation), OLMo-2 1B. Mean off-diagonal = 0.272.

	Care	Fair	Lib	Loy	Auth	Sanc
Care	1.000	0.369	0.288	0.294	0.271	0.321
Fair	0.369	1.000	0.331	0.262	0.286	0.230
Lib	0.288	0.331	1.000	0.200	0.281	0.178
Loy	0.294	0.262	0.200	1.000	0.297	0.238
Auth	0.271	0.286	0.281	0.297	1.000	0.234
Sanc	0.321	0.230	0.178	0.238	0.234	1.000

At layer 0, the within-individualizing mean (0.329) exceeds the within-binding mean (0.256) and the between-group mean (0.258), consistent with the MFT prediction. The care–sanctity pair has higher cosine similarity (0.321) than the care–loyalty pair (0.294), which is not predicted by MFT — sanctity is a binding foundation but shares semantic content with care (both involve purity and harm prevention).

D Permutation Tests for MFT Group Structure

We test whether the six foundation probe directions cluster into the MFT-predicted individualizing (care, fairness, liberty) and binding (loyalty, authority, sanctity) groups. The test statistic is the

Table 5: Cosine similarity matrix at layer 7, OLMo-2 1B. Mean off-diagonal = 0.346.

	Care	Fair	Lib	Loy	Auth	Sanc
Care	1.000	0.398	0.383	0.353	0.283	0.415
Fair	0.398	1.000	0.372	0.329	0.303	0.334
Lib	0.383	0.372	1.000	0.334	0.357	0.303
Loy	0.353	0.329	0.334	1.000	0.389	0.345
Auth	0.283	0.303	0.357	0.389	1.000	0.297
Sanc	0.415	0.334	0.303	0.345	0.297	1.000

Table 6: Cosine similarity matrix at layer 15, OLMo-2 1B. Mean off-diagonal = 0.306.

	Care	Fair	Lib	Loy	Auth	Sanc
Care	1.000	0.406	0.294	0.347	0.324	0.452
Fair	0.406	1.000	0.315	0.267	0.303	0.306
Lib	0.294	0.315	1.000	0.179	0.287	0.243
Loy	0.347	0.267	0.179	1.000	0.360	0.206
Auth	0.324	0.303	0.287	0.360	1.000	0.307
Sanc	0.452	0.306	0.243	0.206	0.307	1.000

difference between mean within-group cosine similarity and mean between-group cosine similarity. We generate the null distribution by permuting group assignments 10,000 times.

Table 7: Permutation test for individualizing/binding group structure across layers, OLMo-2 1B. p-values below 0.05 are bolded.

Layer	Observed statistic	p-value
0	0.035	**0.012**
1	0.012	0.274
2	0.019	0.129
3	0.029	**0.035**
4	0.017	0.147
5	0.015	0.197
6	0.024	0.058
7	0.016	0.195
8	0.013	0.249
9	0.011	0.279
10	0.009	0.332
11	0.007	0.398
12	0.007	0.398
13	0.010	0.330
14	0.014	0.232
15	0.003	0.489

The test reaches significance at layers 0 and 3. With only 20 unique 3–3 partitions of 6 elements, the test has limited power. The dendrogram analysis (Section 4.3) provides complementary qualitative evidence using the full distance matrix rather than the scalar within/between summary.

E Reproducibility

E.1 Hardware and software

All experiments ran on a single MacBook Pro with Apple M4 Pro (24 GB unified memory) using the MPS backend. Software versions: Python 3.14, PyTorch 2.7, Transformers 4.49.

Experiment	Model	Wall time
1–3 (probing + geometry + bootstrap)	OLMo-2 1B	$\sim$5 min
5 (dense vs. MoE geometry)	OLMoE-1B-7B	$\sim$20 min
6 (geometric trajectory)	OLMo-2 1B (20 ckpts)	$\sim$45 min
7 (framework fragility)	OLMo-2 + OLMoE	$\sim$25 min

E.2 Runtime

E.3 Reproducibility notes

Probe training. Linear probes are `nn.Linear(2048, 1)` trained with BCE loss and Adam (lr = 10^{-2}) for 50 epochs. No weight decay, no learning rate schedule. Random seed is not fixed across runs; bootstrap analysis (Appendix~B) quantifies direction stability under resampling.

Probing dataset. The 240-pair minimal-pair dataset (40 per MFT foundation) is deterministic and version-controlled. Dataset generation uses GPT-4 with automated validation gates; the exact dataset is included in the code repository.

Activation collection. Forward passes use `torch.no_grad()`. Activations are mean-pooled across the sequence dimension at each layer. For OLMoE, activations are collected after the expert combination step (post-routing), not from individual experts.

Code availability. All experiment scripts, the probing dataset, and figure generation code are available at `https://github.com/orionr/deepsteer`.